

Test Case: Take Onboarding Survey

TC-SURV-001

Title: Forced redirection to onboarding for a user who hasn't completed it

Preconditions:

- User is logged in with an account where `hasCompletedOnboarding = false`.

Steps:

1. After login, manually navigate to `/dashboard` (or any other protected route).

Expected Result:

- `OnboardingGuard` redirects the user to `/onboarding` instead of allowing access to `/dashboard`.

Priority: High

TC-SURV-002

Title: Step 1 — select weekly study time commitment (auto-advance, no Next/Continue button)

Preconditions:

- User is on `/onboarding`, Step 1 ("Set Your Pace").
- Note: Step 1 has no "Next"/"Continue" button — clicking one of the weekly-hour option cards is itself the action that advances the flow.

Steps:

1. Click one of the weekly hour options (2h / 4h / 6h / 8+h).

Expected Result:

- The selection is registered and the wizard automatically advances to Step 2 ("English Level") immediately — no separate confirmation click is required.

Priority: High

TC-SURV-003

Title: Step 1 — user cannot advance to Step 2 without selecting a weekly hour option

Preconditions:

- User is on `/onboarding`, Step 1 ("Set Your Pace").
- No weekly-hour option has been selected yet.

Steps:

1. On Step 1, do not click any of the weekly hour options (2h / 4h / 6h / 8+h).
2. Try to reach Step 2 by any other means available on the page (e.g. pressing Enter, clicking elsewhere on the card, scrolling) without selecting an option.

Expected Result:

- User remains on Step 1; Step 2 ("English Level") is never displayed.
- There is no "Next"/"Continue"/"Skip" control that would allow bypassing the mandatory selection — advancing is only possible by clicking one of the four pace options.

Priority: Medium

TC-SURV-004

Title: Step 2 — choose "Yes, I know my level" (self-select path) instead of taking the placement test**Preconditions:**

- User has completed Step 1.
- User is on Step 2 ("English Level") showing two option cards: "Yes, I know my level" and "No, I want to take a placement test".

Steps:

1. On Step 2, click the "Yes, I know my level" card.
2. Select a specific CEFR level from the options provided on the next screen.
3. Confirm the selection to proceed.

Expected Result:

- System navigates to a level selection screen after clicking the card.
- After confirming the level, `hasCompletedOnboarding` becomes `true`.
- User is redirected to `/dashboard`.
- A personalized roadmap is generated and displayed based on the manually selected CEFR level.

Priority: High

TC-SURV-005

Title: Step 2 — choose to take the Placement Test instead of self-selecting**Steps:**

1. On Step 2, click the option to take the Placement Test.

Expected Result:

- `PlacementTest.tsx` component launches correctly and starts fetching questions.

Priority: High

TC-SURV-006

Title: Refresh page mid-survey (after Step 1, before Step 2 completion)

Steps:

1. Complete Step 1 (select weekly hours) but do not complete Step 2.
2. Refresh the browser page (F5).

Expected Result:

- Document actual behavior: either Step 1 progress is retained (user resumes at Step 2), or the survey resets to Step 1 — confirm this matches the intended UX design.

Priority: Medium

TC-SURV-007

Title: Use browser back button during the survey

Steps:

1. Start the onboarding survey, complete Step 1, move to Step 2.
2. Click the browser's Back button.

Expected Result:

- No broken UI state, console errors, or crash; app either returns to Step 1 or safely stays on the current step.

Priority: Low

TC-SURV-008

Title: Access to protected routes after completing onboarding

Preconditions:

- User has just completed onboarding (Step 1 + Step 2).

Steps:

1. Try navigating to `/dashboard`, `/roadmap`, `/lessons`.

Expected Result:

- `OnboardingGuard` allows access to all protected routes without redirecting back to `/onboarding`.

Priority: High

TC-SURV-009

Title: Already-onboarded user manually navigates to `/onboarding`

Preconditions:

- User has `hasCompletedOnboarding = true`.

Steps:

1. Manually type `/onboarding` in the browser address bar and navigate.

Expected Result:

- Document actual behavior: system either redirects the user back to `/dashboard`, or allows them to redo onboarding — confirm this matches intended design (currently no explicit guard against re-access was documented, so this may surface a gap).

Priority: Medium

TC-SURV-010

Title: Verify onboarding data is persisted correctly in the database

Preconditions:

- Access to the database (e.g. DBeaver).

Steps:

1. Complete the onboarding survey (Step 1 + Step 2) for a test account.
2. Query the `User` table for that account.

Expected Result:

- `weeklyStudyHours` reflects the selected value from Step 1.
- `hasCompletedOnboarding = true`.
- CEFR level field is populated (self-selected value or placement test result).

Priority: High

UC12 — Take Placement Test

TC-PLACE-001

Title: GET placement test questions does not expose correct answers

Preconditions:

- User is on the Placement Test screen (public endpoint, no login required to view questions).

Steps:

1. Open browser DevTools → Network tab.

2. Trigger `GET /placement-test/questions`.
3. Inspect the JSON response body.

Expected Result:

- Response contains question text and answer options only.
- The `correct_answer` field is NOT present anywhere in the response payload.

Priority: High

TC-PLACE-002

Title: Correct answers are not exposed via Inspect Element / DOM

Steps:

1. Load the Placement Test page.
2. Right-click → Inspect Element on a question, and search the rendered DOM/HTML source for the correct answer.

Expected Result:

- No correct-answer data is embedded in the DOM, inline scripts, or data attributes.

Priority: High

TC-PLACE-003

Title: Answering all questions correctly (100%) assigns the highest CEFR level

Preconditions:

- User has started the Placement Test.

Steps:

1. Answer every question correctly across all levels.
2. Submit the test (`POST /placement-test/submit`).

Expected Result:

- User is assigned the highest CEFR level available in the question bank (e.g. C2).
- Result payload shows 100% (or near-100%) accuracy per level.

Priority: High

TC-PLACE-004

Title: Answering all questions incorrectly (0%) assigns the lowest CEFR level

Steps:

1. Answer every question incorrectly.
2. Submit the test.

Expected Result:

- User is assigned the lowest CEFR level (A1).

Priority: High

TC-PLACE-005

Title: Exactly 70% accuracy at a level — boundary check**Steps:**

1. Answer questions for a given level such that accuracy is exactly 70%.
2. Submit the test.

Expected Result:

- Per the " $\geq 70\%$ " rule, the user advances past that level to the next one (boundary is inclusive).

Priority: Medium

TC-PLACE-006

Title: 69% accuracy at a level — does not advance**Steps:**

1. Answer questions for a given level such that accuracy is just below 70% (e.g. 69%).
2. Submit the test.

Expected Result:

- User does NOT advance past that level; assigned level stays at or below the current level.

Priority: Medium

TC-PLACE-007

Title: Submit the test with unanswered questions**Steps:**

1. Answer only some of the questions, leaving others blank.
2. Submit the test.

Expected Result:

- System handles missing answers gracefully — either treats them as incorrect, or returns a validation error prompting the user to complete all questions (confirm actual behavior).

- No server crash (no HTTP 500).

Priority: Medium

TC-PLACE-008

Title: Submit placement test without a valid Bearer token

Steps:

1. Send **POST** `/placement-test/submit` directly (e.g. via Postman) without an **Authorization** header, or with an expired/invalid token.

Expected Result:

- Request is rejected with HTTP 401 Unauthorized (blocked by **JwtGuard**).

Priority: High

TC-PLACE-009

Title: Successful submission displays result and navigates to dashboard with generated roadmap

Steps:

1. Complete and submit the placement test as a logged-in user.
2. Observe the result screen.
3. Click the "Dashboard" button to proceed.

Expected Result:

- After submission, the system displays a Result page showing the test score and assigned CEFR level.
- User is not auto-redirected; they must explicitly click the button to go to the dashboard.
- Upon clicking, **markOnboardingComplete** is triggered (**hasCompletedOnboarding = true**), user is redirected to `/dashboard`, and a CEFR roadmap is generated based on the assigned level.

Priority: High

TC-PLACE-010

Title: Resubmitting the placement test a second time

Preconditions:

- User has already submitted the placement test once.

Steps:

1. Attempt to call **POST** `/placement-test/submit` again with a new set of answers.

Expected Result:

- Document actual behavior: system either blocks resubmission (test locked after first attempt), allows a retake, or recalculates and overwrites the roadmap — confirm this matches the intended design.

Priority: Low

Test Execution

Test Case ID	Execution Date	Status	Actual Result	Bug ID
TC-SURV-001	2026-08-21	Pass	<code>OnboardingGuard</code> redirected user to <code>/onboarding</code> successfully.	
TC-SURV-002	2026-08-21	Pass	Selection registered and auto-advanced to Step 2 immediately.	
TC-SURV-003	2026-08-21	Pass	Could not advance to Step 2 without clicking a pace option.	
TC-SURV-004	2026-08-21	Fail	API for manual level selection is not connected. User is stuck and cannot complete onboarding.	BUG-004
TC-SURV-005	2026-08-21	Pass	<code>PlacementTest.tsx</code> launched and fetched questions correctly.	
TC-SURV-006	2026-08-21	Pass	Step 1 progress was retained upon page refresh.	
TC-SURV-007	2026-08-21	Pass	Browser back button safely kept user on the current step without crashing.	
TC-SURV-008	2026-08-21	Pass	Accessed protected routes successfully (tested via Placement Test path).	
TC-SURV-009	2026-08-21	Pass	System redirected the onboarded user back to <code>/dashboard</code> .	
TC-SURV-010	2026-08-21	Pass	<code>weeklyStudyHours</code> and CEFR level persisted correctly in DB (via Placement Test path).	
TC-PLACE-001	2026-08-21	Pass	JSON response contained options but no <code>correct_answer</code> fields.	
TC-PLACE-002	2026-08-21	Pass	Inspected DOM; no correct answers were exposed in HTML/attributes.	
TC-PLACE-	2026-08-21	Pass	100% accuracy assigned C2 level correctly.	

Test Case ID	Execution Date	Status	Actual Result	Bug ID
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TC-PLACE-004	2026-08-21	Pass	0% accuracy assigned A1 level correctly.	
TC-PLACE-005	2026-08-21	Pass	Exactly 70% accuracy advanced the user to the next CEFR level.	
TC-PLACE-006	2026-08-21	Pass	69% accuracy kept the user at the current level.	
TC-PLACE-007	2026-08-21	Pass	System handled missing answers as incorrect gracefully without crashing.	
TC-PLACE-008	2026-08-21	Pass	Submission without token was blocked with HTTP 401.	
TC-PLACE-009	2026-08-21	Pass	Result page displayed; clicking Dashboard updated status, redirected, and generated roadmap.	
TC-PLACE-010	2026-08-21	Pass	Resubmission blocked; test locked after first attempt.	